Boolean and LSI Information Retrieval System Evaluation

Measuring Precision and Recall

Introduction

Information Retrieval (IR) systems help users find relevant content in large document collections. Among the oldest and simplest IR models is the Boolean model, which uses logic-based operators like AND and OR to decide whether documents match a user's query.

To go further, we extended the system by adding **Latent Semantic Indexing (LSI)**. Unlike Boolean logic, LSI uses vector math to detect similarity between concepts, not just exact words. This allows for more flexible and meaningful search, especially when users don't know the exact phrasing.

In this study, we evaluated both the Boolean IR system and the new LSI-based system. We compared their results on the same set of queries and documents.

Methodology

Dataset

We worked with a dataset of 42 articles focused on artificial intelligence. These articles cover a variety of AI topics, including large language models, real-world applications, tech innovations, and societal impacts.

The Boolean IR System

Our system, built with Whoosh, goes through three main steps:

1. **Indexing**: Each document is indexed using a schema that includes fields like title, content, and path.
2. **Query Parsing**: User queries are parsed into Boolean logic expressions.
3. **Searching**: Documents are retrieved based on whether they satisfy the Boolean condition.

In this model, documents are treated as sets of terms. A document is considered relevant if it matches the entire Boolean query.

The LSI IR System

The LSI system adds another layer:

- **TF-IDF vectorization**: All documents are transformed into vectors based on word importance.

- **Dimensionality reduction (SVD)**: These vectors are projected into a lower-dimensional space that captures deeper meanings.

- **Similarity scoring**: Queries are compared to documents using cosine similarity.

This method doesn't require exact word matches. It finds documents that are conceptually close to the query.

Evaluation Metrics

To measure the system's performance, we used standard IR metrics:

- **Precision**: How many of the retrieved documents are actually relevant?
  Precision = True Positives / (True Positives + False Positives)
- **Recall**: How many of the relevant documents were successfully retrieved?
  Recall = True Positives / (True Positives + False Negatives)
- **F1-score**: A balance between precision and recall.
  F1 = 2 * (Precision * Recall) / (Precision + Recall)

Queries Used

We tested the system using six queries:

Simple queries:

- "AI", "Nvidia", "open-source"

Complex queries:

- "AI AND Nvidia", "AI OR Google", "AI AND open-source"

Defining Relevance

For the Boolean model, a document was considered relevant if it contained the queried terms.

For the LSI model, relevance was defined by checking if the document was clearly related in topic to the query, even without exact word matches.

Results

Overall Performance

- **Average Precision**: 0.9762
- **Average Recall**: 1.0000
- **Average F1-score**: 0.9877

The system demonstrated excellent performance, retrieving all relevant documents and rarely including irrelevant ones.

Query-by-Query Breakdown (Boolean IR System) :

Query	Precision	Recall	F1-score	Retrieved Docs	Relevant Docs
AI	1.0000	1.0000	1.0000	42	42
Nvidia	1.0000	1.0000	1.0000	13	13
open-source	0.9286	1.0000	0.9630	14	13
AI AND Nvidia	1.0000	1.0000	1.0000	13	13
AI OR Google	1.0000	1.0000	1.0000	42	42
AI AND open-source	0.9286	1.0000	0.9630	14	13
BANANA	0	0	0	0	1

Query-by-Query Breakdown (LSI System) :

Query	Precision 10	Subjective Quality
AI	1.00000	Strong
Nvidia	0.90000	Strong
open-source	0.70000	Good
AI AND Nvidia	0.90000	Strong
AI OR Google	0.90000	Strong
AI AND open-source	0.70000	Good
BANANA	0.00000	Irrevelant

In many cases, LSI surfaced documents that were conceptually related even when the exact query words weren't present. It ranked them by how close their meaning was to the query. This allowed some flexibility in how information was retrieved.

Analysis

Strengths

Boolean

- Perfect recall in all relevant queries.
- High precision overall.
- Predictable and consistent behavior.

LSI

- Captures meaning, not just keywords.
- Finds related content even if words don’t exactly match.
- Useful for vague or exploratory queries.

Weaknesses

Boolean

- Too strict: misses anything that doesn’t match exactly.
- Requires users to think in logic expressions.

LSI

- Can include off-topic results, especially for short or vague queries.
- Less transparent: harder to understand why some documents were chosen.

Conclusion

Our evaluation shows that the Boolean IR system is highly effective for retrieving relevant documents when the query terms are well-defined. It achieved perfect recall and near-perfect precision across all tested cases.

The LSI system, while slightly less precise, adds value by retrieving semantically related documents. This makes it useful for cases where users want more flexibility or don’t know the exact terms to use.

Both systems have strengths. Boolean is better for exact, targeted queries. LSI shines when exploring ideas, themes, or loosely defined needs.

Future enhancements could include:

- Mixing Boolean filtering with LSI ranking
- Using NLP tools to detect synonyms or related terms
- Testing on a larger and more diverse dataset